



TECHNICAL NOTE

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Customer Lifetime Value

Companies with recurring revenues from customers often use customer relationship management (CRM) software to identify profitable customer segments and sub-segments and calculate the customer lifetime value of each type of customer. However, complex software applications are not always necessary for such calculations, as they simply require estimates of certain customer metrics, including the cost of contact, likelihood of purchase, purchase margin, and attrition (or conversely, retention) rate. Using this data, a company can make important calculations about its target customers and the profit potential of various marketing efforts.

Customer Acquisition Costs

Consider Matsuzaka Steak,¹ a Japanese online specialty retailer that grew out of a small restaurant/butcher shop in the Chiba prefecture of the greater Tokyo area. The company markets gourmet steaks and chops from its e-commerce site at www.matsuzaka-steak.com. Matsuzaka Steak specializes in Matsuzaka beef, a black-haired wagyu brand that in the minds of many consumers rivals the quality of its more famous cousins, Kobe and Yonezawa. Matsuzaka's exquisite marbling and butter-like tenderness place it at an extraordinarily high price point, often in excess of ¥50,000² per kilogram. As a result, Matsuzaka Steak's online business caters to two primary use cases: gourmet food enthusiasts (or "foodies") who purchase regularly for their own consumption, and less frequent purchasers who visit www.matsuzaka-steak.com primarily to purchase gifts around the holidays. Matsuzaka Steak ships the products ordered online to addresses in Japan in sealed containers filled with dry ice to preserve the meat's freshness.

Imagine that Matsuzaka Steak is contemplating the purchase of a list from a list broker for ¥180 per name. The company's intent is to send each person on the list a full-color catalog, which costs the company ¥48 to produce and ¥30 to mail, every month for a year; those who do not purchase within one year will be removed from the mailing list. The company's internal marketing team reports that such lists generally result in a response rate of approximately 7.5%—that is, 7.5% of customers who receive an unsolicited catalog in this manner will make a

¹ Although the background information on Matsuzaka Steak is accurate, none of the financial information should be interpreted as representative of the company's actual business; the authors fabricated it to illustrate customer lifetime value concepts in action rather than to depict an accurate picture of the economics of a mail-order food products business.

² Although not relevant to the calculations, the Japanese yen (¥) was trading at about 120 per U.S. dollar on October 16, 2015, making Matsuzaka beef approximately \$200 per pound.

purchase. Customers who do not identify as foodies will exhibit a response rate of only 5%. One can therefore calculate Matsuzaka Steak's cost of acquiring a customer (whether foodie or non-foodie) as follows:³

$$\text{Acquisition Cost} = \frac{\text{Cost to Contact}}{\text{Response Rate}}$$

For Matsuzaka Steak, the cost of acquiring a foodie customer by mailing catalogs to this list would be:

$$\text{Acquisition Cost} = \frac{¥180 + 12 * (¥48 + ¥30)}{0.075} = ¥14,880$$

Acquiring a non-foodie customer would cost:

$$\text{Acquisition Cost} = \frac{12 * (¥48 + ¥30)}{0.05} = ¥18,720$$

In this case, even though the cost of the broker's list makes contacting the foodie customers almost 20% more expensive (¥1,116 versus ¥936), the increased response rate more than justifies this added cost and Matsuzaka would go forward with purchasing the list.

The company must recognize that lists featuring attractive foodie customers are necessarily finite; there is a limited number of gourmet food aficionados just as there is a limited number of model car enthusiasts, former Olympic athletes, or any other target customer segment. As a result, companies eventually must purchase lists of customers with potentially lower response rates and must account for the lower response rates when deciding whether to purchase the lists.⁴

Customer Break-Even

Once Matsuzaka Steak knows how much it costs to acquire customers, it can calculate how long it takes to recoup the costs through customer purchases.

Suppose that Matsuzaka Steak sends out catalogs each month to all customers who have made purchases. Foodie customers purchase more frequently than non-foodies, making three purchases per year with an average order size of ¥6,000; non-foodies make one large gift purchase annually with an average order size of ¥15,000. Foodies also tend to remain customers longer, with an annual retention rate (the percentage of customers who will continue to make purchases the next year) of 70% versus a retention rate of 60% for non-foodies.

Lastly, because they view gourmet products like Matsuzaka Steak as staples, foodie customers tend to purchase slightly less expensive products than their non-foodie counterparts,

³ Some of the information in this section comes from Elie Ofek, "Customer Profitability and Lifetime Value," white paper #9-503-019 (Harvard Business School, August 7, 2002).

⁴ Ibid.

who view Matsuzaka Steak as special-occasion products on which to splurge. This behavior results in a margin to Matsuzaka Steak of 60% for non-foodies versus a margin of 50% for foodies.

With this information, we can build tables that show how long it will take to break even on both types of customers.

Foodie Customers	Year 1	Year 2	Year 3	Year 4	Year 5
Annual orders	3	3	3	3	3
Order size	¥6,000	¥6,000	¥6,000	¥6,000	¥6,000
Margin	50%	50%	50%	50%	50%
Annual margin	¥9,000	¥9,000	¥9,000	¥9,000	¥9,000
Survival rate ^a	100%	70.0%	49.0%	34.3%	24.0%
Customer acquisition cost ^b	¥14,880				
Catalog mailing cost		¥936	¥936	¥936	¥936
Annual profit	-¥5,880	¥5,645	¥3,951	¥2,766	¥1,936
Cumulative profit ^c	-¥5,880	-¥235	¥3,716	¥6,482	¥8,418

^a The n-year survival rate is calculated as $(\text{retention rate})^{n-1}$. For example, the four-year survival rate for foodie customers will be $(0.70)^3$, or approximately 34%, while non-foodie customers will have a four-year survival rate of $(0.60)^3$, or approximately 21%.

^b Note that the customer acquisition cost occurs upfront; for each catalog recipient who becomes a customer, the cost of sending the catalog to all recipients who did not become customers is included. This cost only occurs in year 1.

^c Savvy readers will note that cash flows should be discounted for the time value of money, and that one should account for inflation in growing both costs and revenues. We have not done so here simply because we prefer to focus instead on the baseline calculations regarding customer profitability; the necessity of discounting cash flows is covered in the Customer Lifetime Value section below. Discounting cash flows will lengthen the time to break even when customer acquisition involves an upfront expense.

Matsuzaka Steak will break even on foodie customers in year 3. In year 1, the company loses ¥5,880 because the customer acquisition cost exceeds the annual margin. In year 2, 30% of customers have succumbed to attrition and ceased to be customers. The company makes an additional ¥5,645 on each remaining customer, which brings the cumulative loss on each customer to ¥235. In year 3, the customer relationship finally turns profitable. Although attrition reduces the annual profit per customer in the out years, the value of the relationship grows over time as the remaining customers continue to make purchases.

An analysis of non-foodie customers reveals quite a different picture.

Non-Foodie Customers	Year 1	Year 2	Year 3	Year 4	Year 5
Annual orders	1	1	1	1	1
Order size	¥15,000	¥15,000	¥15,000	¥15,000	¥15,000
Margin	60%	60%	60%	60%	60%
Annual margin	¥9,000	¥9,000	¥9,000	¥9,000	¥9,000
Survival rate	100%	60%	36%	22%	13%
Customer acquisition cost	¥18,720				
Catalog mailing cost		¥936	¥936	¥936	¥936
Annual profit	-¥9,720	¥4,838	¥2,903	¥1,742	¥1,045
Cumulative profit	-¥9,720	-¥4,882	-¥1,979	-¥237	¥808

Matsuzaka Steak will break even on non-foodie customers in year 5. The longer break-even is due in part to the higher customer acquisition cost, and in part to the higher attrition (and lower

survival) rate. It is noteworthy that the annual dollar margin on sales to foodies and non-foodies alike is identical at ¥9,000; the higher order frequency of foodies is offset by the higher margin on non-foodie purchases. Foodie customers therefore become profitable more quickly not because they buy more, but rather because they stay customers longer and are less expensive to acquire because of their higher response rate.

It is important to note that these calculations are highly sensitive to the assumptions used. For example, if the non-foodie survival rate drops just five percentage points—from 60% to 55%—the non-foodie customer will not become profitable until year 9, a full four years later. The cumulative profitability will also decrease significantly. The longer it takes for a customer to break even, the less likely it is to occur; new competitors enter, markets change in size, and customer behavior evolves, all of which means that today's reasonable assumptions are unlikely to stay reasonable long into the future.

Customer Lifetime Value

Once Matsuzaka Steak knows what it costs to acquire a customer and how long it will take that customer to become profitable, it can determine the lifetime value of a customer. Customer lifetime value is equal to the net present value (NPV) of all the cash flows associated with a customer over the life of its relationship with the company.

When calculating customer lifetime value each cash flow must be discounted to represent the time value of money.⁵ The customer lifetime value tables below for Matsuzaka Steak assume a 10% discount rate. They also reflect the simplifying assumption that once the NPV of annual profit dips below ¥120, complete customer attrition has effectively occurred and customer lifetime has ended.

⁵ A cash flow is discounted by dividing it by $(1 + \text{discount rate})^n$, where n equals the number of years in the future until that cash flow occurs. The sum of discounted cash flows is the NPV.

Foodie Customers	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10	Year 11
Annual margin	¥9,000	¥9,000	¥9,000	¥9,000	¥9,000	¥9,000	¥9,000	¥9,000	¥9,000	¥9,000	¥9,000
Survival rate	100%	70%	49%	34.3%	24%	16.8%	11.8%	8.2%	5.8%	4%	2.8%
Customer acquisition cost	¥14,880										
Catalog mailing cost		¥936	¥936	¥936	¥936	¥936	¥936	¥936	¥936	¥936	¥936
Annual profit	-¥5,880	¥5,645	¥3,951	¥2,766	¥1,936	¥1,355	¥949	¥664	¥465	¥325	¥228
NPV of annual profit	-¥5,880	¥5,132	¥3,266	¥2,078	¥1,322	¥842	¥536	¥341	¥217	¥138	¥88
Cumulative profit	-¥5,880	-¥235	¥3,716	¥6,482	¥8,418	¥9,774	¥10,722	¥11,386	¥11,851	¥12,177	¥12,404
NPV of cumulative profit	-¥5,880	-¥748	¥2,517	¥4,595	¥5,918	¥6,759	¥7,295	¥7,636	¥7,852	¥7,990	¥8,078

The NPV of annual profit for foodies falls below ¥120 in year 11, at which point a 0% survival rate is assumed and the calculation stops. The result is a customer lifetime value of ¥8,078 for Matsuzaka Steak's foodie customers.

Non-Foodie Customers	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
Annual margin	¥9,000	¥9,000	¥9,000	¥9,000	¥9,000	¥9,000	¥9,000	¥9,000
Survival rate	100%	60%	36%	22%	13%	8%	5%	3%
Customer acquisition cost	¥18,720							
Catalog mailing cost		¥936	¥936	¥936	¥936	¥936	¥936	¥936
Annual profit	-¥9,720	¥4,838	¥2,903	¥1,742	¥1,045	¥627	¥376	¥226
NPV of annual profit	-¥9,720	¥4,399	¥2,399	¥1,309	¥714	¥389	¥212	¥116
Cumulative profit	-¥9,720	-¥4,882	-¥1,979	-¥237	¥808	¥1,435	¥1,812	¥2,037
NPV of cumulative profit ^a	-¥9,720	-¥5,321	-¥2,922	-¥1,614	-¥900	-¥510	-¥298	-¥182

^a The NPV of cumulative profit function is an asymptotic function without the assumption that 100% of customers attrite when the NPV falls below ¥120. For foodies, it is asymptotic approaching ¥8,231 and for non-foodies it approaches -¥43.

Non-foodies, on the other hand, never become profitable once future cash flows are discounted, in large part due to high attrition rates that shorten customer lifetime to only eight years. This result suggests that non-foodies with the assumed characteristics would be an unprofitable customer segment for Matsuzaka Steak to target.